INSTITUTE SEMINAR

Besov Spaces for Partial Differential Equations

Littlewood–Paley decomposition, nonlinear estimates, and critical regularity



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Institute seminar • 13 October 2026 • 9:00 AM

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Starting from Sobolev spaces

Measure derivatives in $L^p(\mathbb{R}^d)$

For an integer $k \geq 0$ and $1 \leq p \leq \infty$,

$$W^{k,p} = \{u \in L^p : \partial^\alpha u \in L^p \text{ for } |\alpha| \leq k\}, \qquad \|u\|_{W^{k,p}} = \sum_{|\alpha| \leq k} \|\partial^\alpha u\|_p.$$

A familiar starting point

Differentiability and integrability describe the size and regularity of a solution.

For $p = 2$, integer-order Sobolev spaces are denoted by H^k.

The question for this talk

How can we measure regularity **one frequency scale at a time**?

We will also choose how to sum the contributions of different scales.

First construct the frequency tools. Then use them to estimate PDEs.

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The Fourier transform and frequency

Our convention on $\mathbb{R}^d$

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} e^{-ix\cdot\xi} f(x)\,dx, \qquad f(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{ix\cdot\xi} \hat{f}(\xi)\,d\xi.$$

Start with Schwartz functions, then extend to tempered distributions.

What does ξ measure?

The wave $e^{ix\cdot\xi}$ oscillates in direction ξ.

Its spatial scale is proportional to $|\xi|^{-1}$. Large $|\xi|$ means rapid oscillation.

Derivatives become multipliers

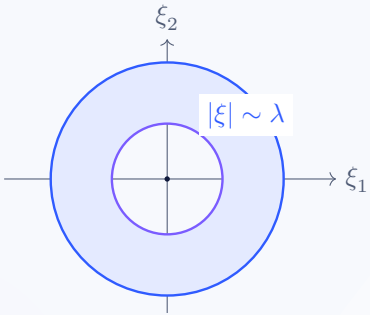
$$\widehat{\partial_k f} = i\xi_k \hat{f}, \qquad \widehat{-\Delta f} = |\xi|^2 \hat{f}.$$

Frequency localization makes the size of these multipliers predictable.

The exponentials illustrate oscillation. They are not finite-p functions in $L^p(\mathbb{R}^d)$.

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Functions localized in frequency



an annulus avoids the origin

Two support conditions

Ball: $\operatorname{supp} \hat{f} \subset \{|\xi| \leq C_0\lambda\}$.

Annulus: $\operatorname{supp} \hat{f} \subset \{c_0\lambda \leq |\xi| \leq C_0\lambda\}$, with $c_0 > 0$.

The expected size

On the annulus, one derivative has size λ and two have size λ^2.

Bernstein makes this precise in L^p.

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Bernstein inequalities

Ball support: $\operatorname{supp} \widehat{f} \subset \{|\xi| \leq C_0 \lambda\}$

For $1 \leq p \leq q \leq \infty$ and an integer $k \geq 0$,

$$\boxed{\|\nabla^k f\|_{L^q} \leq C_k \lambda^{k+d(1/p-1/q)} \|f\|_{L^p}.}$$

Annular support: $c_0 \lambda \leq |\xi| \leq C_0 \lambda$

For $1 \leq p \leq \infty$,

$$\boxed{\|\nabla^k f\|_{L^p} \simeq_k \lambda^k \|f\|_{L^p}.}$$

The collection ∇^k contains **all** derivatives of order k.

The constants depend on the support geometry and exponents, but are independent of λ and f.

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Why Bernstein works

A smooth cutoff equal to one on the Fourier support

Choose $\psi \in C_c^\infty$ and write $K = \mathcal{F}^{-1}\psi$. Then

$$f = K_\lambda * f, \qquad K_\lambda(x) = \lambda^d K(\lambda x).$$

If $1 + 1/q = 1/p + 1/a$, Young's inequality gives

$$\|\partial^\alpha f\|_q \le \|\partial^\alpha K_\lambda\|_a \|f\|_p = \lambda^{|\alpha| + d(1-1/a)} \|\partial^\alpha K\|_a \|f\|_p.$$

Why the annulus also gives a reverse bound

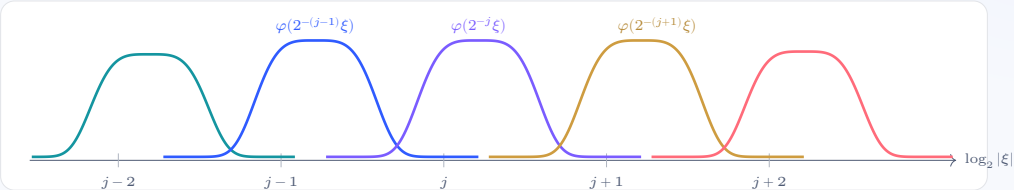
Away from $\xi = 0$, smooth localized multipliers recover f from its full gradient. Their kernels have L^1 size $O(\lambda^{-1})$. Iteration gives the order-k result.

The support information becomes a rescaling of an integrable kernel.

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A dyadic partition of unity

Choose a smooth radial $\chi = 1$ on $|\xi| \le \frac{3}{4}$, supported in $|\xi| \le \frac{4}{3}$, and set $\varphi(\xi) = \chi(\xi/2) - \chi(\xi)$.



Inhomogeneous

$$\chi(\xi) + \sum_{j \ge 0} \varphi(2^{-j}\xi) = 1, \qquad \xi \in \mathbb{R}^d.$$

Homogeneous

$$\sum_{j \in \mathbb{Z}} \varphi(2^{-j}\xi) = 1, \qquad \xi \neq 0.$$

The curves illustrate overlapping bands. The displayed identities specify the exact partition.

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Littlewood–Paley blocks

Homogeneous annular blocks: $j \in \mathbb{Z}$

$$\boxed{\dot{\Delta}_j u = \mathcal{F}^{-1}[\varphi(2^{-j}\xi)\hat{u}(\xi)]} \qquad \operatorname{supp} \widehat{\dot{\Delta}_j u} \subset \{\tfrac{3}{4}2^j \le |\xi| \le \tfrac{8}{3}2^j\}.$$

$$\dot{\Delta}_j u = h_j * u, \qquad h = \mathcal{F}^{-1}\varphi, \qquad h_j(x) = 2^{jd}h(2^j x).$$

Homogeneous low-frequency cutoff

$$\boxed{\dot{S}_j u = \mathcal{F}^{-1}[\chi(2^{-j}\xi)\hat{u}(\xi)]} \qquad \operatorname{supp} \widehat{\dot{S}_j u} \subset \{|\xi| \le \tfrac{4}{3}2^j\}.$$

The kernels give uniform L^p bounds, $1 \le p \le \infty$.
Only neighboring annuli overlap: $\dot{\Delta}_j \dot{\Delta}_k = 0$ if $|j-k| \ge 2$.

These are smooth frequency filters, not orthogonal projections. Dots always indicate homogeneous operators.

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Reconstructing the function

The telescoping identity

$$\sum_{j=m}^{n} \dot{\Delta}_j u = \dot{S}_{n+1} u - \dot{S}_m u.$$

Every polynomial has zero annular blocks. We need a low-frequency normalization.

Our homogeneous convention

$$\mathcal{S}'_h = \left\{ u \in \mathcal{S}' : \lim_{m \to -\infty} \|\dot{S}_m u\|_\infty = 0 \right\}.$$

For $u \in \mathcal{S}'_h$,

$$\boxed{u = \sum_{j \in \mathbb{Z}} \dot{\Delta}_j u \quad \text{in } \mathcal{S}',} \qquad \dot{S}_j u = \sum_{k<j} \dot{\Delta}_k u.$$

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Homogeneous Besov spaces

For $s \in \mathbb{R}$, $1 \le p, r \le \infty$, and $u \in \mathcal{S}'_h$

$$\|u\|_{\dot{B}^s_{p,r}} = \left\| \left(2^{js} \|\dot{\Delta}_j u\|_{L^p} \right)_{j \in \mathbb{Z}} \right\|_{\ell^r}$$

$$\dot{B}^s_{p,r} = \{ u \in \mathcal{S}'_h : \|u\|_{\dot{B}^s_{p,r}} < \infty \}.$$

s: regularity
The weight 2^{js} records the number of derivatives.

p: spatial size
Each block is measured in $L^p(\mathbb{R}^d)$.

r: sum of scales
Use the ℓ^r norm, or the supremum if $r = \infty$.

This realization is complete for $s < d/p$, or $s = d/p$ with $r = 1$. Higher smoothing norms are paired with a lower normalized norm. See backup B1.

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Inhomogeneous Besov spaces

One low block, followed by the nonnegative annuli

$$\Delta_{-1}u = \mathcal{F}^{-1}[\chi\hat{u}], \qquad \Delta_j u = \dot{\Delta}_j u \ (j \geq 0), \qquad u = \sum_{j\geq -1} \Delta_j u \quad \text{in } \mathcal{S}'.$$

$$\boxed{\|u\|_{B^s_{p,r}} = \left\| \left(2^{js}\|\Delta_j u\|_p\right)_{j\geq -1} \right\|_{\ell^r}}, \qquad B^s_{p,r} = \{u \in \mathcal{S}' : \|u\|_{B^s_{p,r}} < \infty\}.$$

The distinction is at low frequency

Inhomogeneous: collect the origin in Δ_{-1}. No polynomial ambiguity.
Homogeneous: keep all integer scales and use the normalization in $\mathcal{S}'_h$.

The familiar Sobolev case is $B^s_{2,2} = H^s$, with equivalent norms.

From now on, every PDE estimate uses $\dot{\Delta}_j$ and $\dot{B}^s_{p,r}$.

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Differentiation and scaling

Bernstein applies to each homogeneous block

$$\|\nabla \dot{\Delta}_j u\|_p \simeq 2^j \|\dot{\Delta}_j u\|_p \quad \Longrightarrow \quad \boxed{\|\nabla u\|_{\dot{B}^{s-1}_{p,r}} \simeq \|u\|_{\dot{B}^{s}_{p,r}}.}$$

The equivalence uses the full gradient.

Changing the spatial scale

For $u_\lambda(x) = u(\lambda x)$,

$$\boxed{\|u_\lambda\|_{\dot{B}^{s}_{p,r}} \asymp \lambda^{s-d/p} \|u\|_{\dot{B}^{s}_{p,r}}.}$$

For dyadic $\lambda = 2^m$, the identity is exact with the fixed dyadic norm.
For arbitrary $\lambda > 0$, the comparison constants are uniform.

Differentiation changes the weight. Scaling shifts the frequency index.

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Embeddings and the critical endpoint

Spatial integrability: $1 \le p_1 \le p_2 \le \infty$

$$\dot{B}^{s}_{p_1,r} \hookrightarrow \dot{B}^{s-d(1/p_1-1/p_2)}_{p_2,r}.$$

Apply Bernstein to each block. The exponent $s - d/p$ stays unchanged.

The endpoint $\dot{B}^{d/p}_{p,1} \hookrightarrow L^\infty$

$$\|u\|_\infty \le \sum_j \|\dot{\Delta}_j u\|_\infty \lesssim \sum_j 2^{jd/p} \|\dot{\Delta}_j u\|_p = \|u\|_{\dot{B}^{d/p}_{p,1}}.$$

The triangle inequality requires the sum of all block sizes. This is why $r = 1$ matters.

For transport: $\|\nabla u\|_\infty \lesssim \|u\|_{\dot{B}^{d/p+1}_{p,1}}$.

The endpoint generally fails for $r > 1$. A whole-space counterexample is in backup B2.

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The linear heat equation

$$\partial_t u - \nu \Delta u = f, \qquad u(0) = u_0, \qquad \nu > 0.$$

In Fourier variables

$$\partial_t \hat{u} + \nu |\xi|^2 \hat{u} = \hat{f}.$$

With $f = 0$, $\hat{u}(t, \xi) = e^{-\nu t |\xi|^2} \hat{u}_0(\xi)$.

Duhamel's formula

$$u(t) = e^{\nu t \Delta} u_0 + \int_0^t e^{\nu (t-\tau) \Delta} f(\tau)\, d\tau.$$

Set $u_j = \dot{\Delta}_j u$ and $f_j = \dot{\Delta}_j f$. The blocks commute with the heat operator:

$$\partial_t u_j - \nu \Delta u_j = f_j.$$

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Heat estimates on each frequency block

First take $f = 0$, with $1 \le p \le \infty$

$$\|u_j(t)\|_p \le Ce^{-c\nu t 2^{2j}} \|u_{0,j}\|_p.$$

For $p = 2$, use Plancherel. For general p, localize the multiplier and bound its kernel in L^1.

Two consequences of the same block estimate

$$\|u(t)\|_{\dot{B}^s_{p,r}} \lesssim \|u_0\|_{\dot{B}^s_{p,r}},$$

$$\nu 2^{j(s+2)} \|u_j\|_{L^1(0,T;L^p)} \lesssim 2^{js} \|u_{0,j}\|_p.$$

First sum at fixed time, or integrate each block before summation.

Time integration contributes $(\nu 2^{2j})^{-1}$, paying for two derivatives.

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The linear transport equation

$$\partial_t a + v \cdot \nabla a = f, \qquad a(0) = a_0.$$

A prescribed smooth velocity v

Along its flow X, defined by $\partial_t X(t,x) = v(t, X(t,x))$ and $X(0,x) = x$,

$$\frac{d}{dt} a(t, X(t,x)) = f(t, X(t,x)).$$

The basic L^p estimate

$$\frac{d}{dt} \|a(t)\|_p \le \|f(t)\|_p + \frac{1}{p} \|\operatorname{div} v(t)\|_\infty \|a(t)\|_p.$$

Interpret this in integrated form. At $p = \infty$, use $1/p = 0$.
If $\operatorname{div} v = 0$ and $f = 0$, the L^p norm is conserved.

Transport supplies no heat smoothing.

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Combining time and spatial regularity

The ordinary Bochner norm

Set $b_j(t) = 2^{js}\|\dot{\Delta}_j u(t)\|_p$ for $0 < t < T$. Then

$$\boxed{\|u\|_{L^\rho_T(\dot{B}^s_{p,r})} = \big\|\|(b_j(t))_j\|_{\ell^r}\big\|_{L^\rho_t}.}$$

Sum the frequencies at each time, then integrate in time.

What did the heat calculation produce?

We took a time norm of each block, then wanted to sum in frequency.

Exchanging these operations can change the norm.

Ordinary mixed Lebesgue estimates work away from the endpoints.
The dyadic argument also gives useful endpoint estimates.

We write $L^\rho_T = L^\rho(0,T)$ and keep p for the spatial exponent.

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Chemin–Lerner spaces

Take the time norm before the frequency sum

$$\|u\|_{\widetilde{L}^{\rho}_{T}(\dot{B}^{s}_{p,r})} = \left\| \left(2^{js} \|\dot{\Delta}_j u\|_{L^{\rho}(0,T;L^{p})} \right)_{j\in\mathbb{Z}} \right\|_{\ell^{r}}.$$

Here $1 \le p, r, \rho \le \infty$. An infinite exponent means the corresponding supremum.

Bochner

$$L^{\rho}_{t}(\ell^{r}_{j}).$$

Measure all scales at the same time.

Chemin–Lerner

$$\ell^{r}_{j}(L^{\rho}_{t}).$$

Measure each scale over the whole time interval.

This norm records exactly the order of the heat proof.

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Why the order of the norms matters

Minkowski comparison, with $\mathcal{B} = \dot{B}^s_{p,r}$

$$\rho \le r: \quad \|u\|_{\widetilde{L}^\rho_T\mathcal{B}} \le \|u\|_{L^\rho_T\mathcal{B}}, \quad L^\rho_T\mathcal{B} \hookrightarrow \widetilde{L}^\rho_T\mathcal{B},$$
$$r \le \rho: \quad \|u\|_{L^\rho_T\mathcal{B}} \le \|u\|_{\widetilde{L}^\rho_T\mathcal{B}}, \quad \widetilde{L}^\rho_T\mathcal{B} \hookrightarrow L^\rho_T\mathcal{B}.$$

If $r = \rho$, the norms agree.

Two amplitudes peak at different times: $r = 1$, $\rho = \infty$

Let $b_1 = \mathbf{1}_{(0,1)}$ and $b_2 = \mathbf{1}_{(1,2)}$.

$$\sup_t (b_1 + b_2) = 1, \qquad \sup_t b_1 + \sup_t b_2 = 2.$$

The Chemin–Lerner norm remembers both peaks.

$$\widetilde{L}^\infty_T \dot{B}^s_{p,1} \hookrightarrow L^\infty_T \dot{B}^s_{p,1}, \qquad \widetilde{L}^1_T \dot{B}^s_{p,1} = L^1_T \dot{B}^s_{p,1}.$$

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The time-dependent heat estimate

Take L^ρ in time at each scale

$$\|u_j\|_{L^\rho_T L^p} \lesssim (\nu 2^{2j})^{-1/\rho} \left(\|u_{0,j}\|_p + \|f_j\|_{L^1_T L^p} \right).$$

Multiply by $\nu^{1/\rho} 2^{j(s+2/\rho)}$, then sum in ℓ^r.

$$\boxed{\nu^{1/\rho} \|u\|_{\widetilde{L}^\rho_T(\dot{B}^{s+2/\rho}_{p,r})} \lesssim \|u_0\|_{\dot{B}^s_{p,r}} + \|f\|_{\widetilde{L}^1_T(\dot{B}^s_{p,r})}.}$$

$1 \le p, r, \rho \le \infty$, with $1/\infty = 0$.
The constant is independent of T and $\nu > 0$.

$\rho = \infty$: retain s derivatives. $\rho = 1$: gain two derivatives in time integral.

These norms use spatial blocks. They are different from Besov regularity in the time variable.

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Introducing nonlinear convection

Two typical terms

$$\partial_t u - \nu\Delta u = -u\cdot\nabla u + f, \qquad \partial_t a + u\cdot\nabla a = f.$$

The same velocity can be an unknown and a transporting field.

Multiplication mixes frequencies

$$\widehat{fg}(\xi) = (2\pi)^{-d}\int \hat{f}(\eta)\hat{g}(\xi-\eta)\,d\eta, \qquad \dot{\Delta}_j(fg) \neq (\dot{\Delta}_j f)(\dot{\Delta}_j g).$$

Different pairs of input frequencies can produce the same output frequency.

We need a product estimate and a way to track the derivative in convection.

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Bony's paraproduct decomposition

$$\boxed{uv = T_u v + T_v u + R(u,v).}$$

Low times high, and its symmetric partner

$$T_u v = \sum_{j\in\mathbb{Z}} \dot{S}_{j-1}u\,\dot{\Delta}_j v, \qquad \dot{S}_{j-1}u = \sum_{k\le j-2} \dot{\Delta}_k u.$$

$T_v u$ interchanges the two factors.

Comparable frequencies

$$R(u,v) = \sum_j \dot{\Delta}_j u\,\widetilde{\dot{\Delta}}_j v, \qquad \widetilde{\dot{\Delta}}_j = \dot{\Delta}_{j-1} + \dot{\Delta}_j + \dot{\Delta}_{j+1}.$$

First an identity for Schwartz functions. Extension to rough distributions requires convergence of the terms.

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Frequency interactions in a product

Separated scales

$$k \le j-2: \quad \dot{\Delta}_k u \, \dot{\Delta}_j v.$$

The output lies in an annulus of size 2^j.

A low factor changes the amplitude of a fast oscillation.

Comparable scales

$$|j-k| \le 1: \quad \dot{\Delta}_j u \, \dot{\Delta}_k v.$$

The output lies in a ball of size 2^j.

Cancellation can create arbitrarily low frequencies.

A frequency calculation

$$e^{i2^j x} \, e^{-i2^j x} = 1.$$

High input frequencies can create zero output frequency.

The exponentials illustrate frequency support. They are not $L^p(\mathbb{R}^d)$ examples for finite p.

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Product laws

Tame estimate: $s > 0$, $1 \le p < \infty$, $1 \le r \le \infty$

For $f, g \in L^\infty \cap \dot{B}^s_{p,r}$,

$$\|fg\|_{\dot{B}^s_{p,r}} \lesssim \|f\|_\infty \|g\|_{\dot{B}^s_{p,r}} + \|g\|_\infty \|f\|_{\dot{B}^s_{p,r}}.$$

Together with the endpoint embedding, $\dot{B}^{d/p}_{p,1}$ is an algebra for finite p.

The multiplier estimate for our nonlinear example

For $-d/2 < \sigma \le d/2$,

$$\boxed{\|fg\|_{\dot{B}^\sigma_{2,1}} \lesssim \|f\|_{\dot{B}^{d/2}_{2,1}} \|g\|_{\dot{B}^\sigma_{2,1}}.}$$

This also covers $\sigma = 0$ and negative σ in the displayed range.

All spaces here are homogeneous with our fixed normalization. The second bound follows from Bony and Bernstein; backup B8 gives the summation.

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Short proof ideas for the product laws

Paraproduct: separated frequencies

$$\|\dot{\Delta}_j T_f g\|_p \lesssim \|f\|_\infty \sum_{|k-j|\le N_0} \|\dot{\Delta}_k g\|_p.$$

Only a fixed number of output shells interact. Weight by 2^{js} and sum.

Remainder: comparable inputs, possibly low output

$$2^{js}\|\dot{\Delta}_j R(f,g)\|_p \lesssim \|f\|_\infty \sum_{k\ge j-N_0} 2^{(j-k)s}\, 2^{ks}\|\widetilde{\dot{\Delta}}_k g\|_p.$$

The geometric kernel is summable when $s > 0$.

For the multiplier law, combine $L^2 \cdot L^2 \subset L^1$ with Bernstein. The high–high sum requires $\sigma + d/2 > 0$. Here N_0 is a fixed overlap constant.

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The derivative cost of convection

One concrete critical example: $d \geq 2$, $p = 2$, $r = 1$

Set $\sigma = d/2 - 1$. The scaling $u_\lambda(t,x) = \lambda u(\lambda^2 t, \lambda x)$ preserves the $\dot{B}^\sigma_{2,1}$ norm up to equivalence.

$$\|u \cdot \nabla u\|_{\dot{B}^\sigma_{2,1}} \lesssim \|u\|_{\dot{B}^{d/2}_{2,1}} \|\nabla u\|_{\dot{B}^\sigma_{2,1}} \lesssim \|u\|^2_{\dot{B}^{\sigma+1}_{2,1}}.$$

Interpolation connects this to heat smoothing

$$\|u\|^2_{\dot{B}^{\sigma+1}_{2,1}} \leq \|u\|_{\dot{B}^\sigma_{2,1}} \|u\|_{\dot{B}^{\sigma+2}_{2,1}}.$$

Hence

$$\boxed{\|u \cdot \nabla u\|_{L^1_T \dot{B}^\sigma_{2,1}} \lesssim \|u\|_{L^\infty_T \dot{B}^\sigma_{2,1}} \|u\|_{L^1_T \dot{B}^{\sigma+2}_{2,1}}.}$$

The interpolation inequality is Cauchy–Schwarz applied to the weighted dyadic sum. Two extra derivatives enter with time integrability.

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Closing a nonlinear heat estimate

Convective heat model: $\partial_t u - \nu\Delta u = -u\cdot\nabla u + f$

Continue with $d \geq 2$ and $\sigma = d/2 - 1$. Define

$$X_T = \|u\|_{\widetilde{L}^\infty_T \dot{B}^\sigma_{2,1}} + \nu\|u\|_{L^1_T \dot{B}^{\sigma+2}_{2,1}}, \qquad D_T = \|u_0\|_{\dot{B}^\sigma_{2,1}} + \|f\|_{L^1_T \dot{B}^\sigma_{2,1}}.$$

The linear estimate plus the product bound

$$X_T \leq C\left(D_T + \|u\cdot\nabla u\|_{L^1_T \dot{B}^\sigma_{2,1}}\right) \leq CD_T + \frac{C}{\nu}X_T^2.$$

$$\boxed{D_T \text{ sufficiently small relative to } \nu \quad \Longrightarrow \quad X_T \lesssim D_T}$$

by a continuity argument for smooth solutions.

The estimate closes because smoothing controls the convective derivative.

This is an a priori mechanism. Constructing a solution and proving uniqueness require additional steps.

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Transport and commutator cancellation

Localize $\partial_t a + v \cdot \nabla a = f$

$$\partial_t \dot{\Delta}_j a + v \cdot \nabla \dot{\Delta}_j a = \dot{\Delta}_j f + C_j, \qquad C_j = [v \cdot \nabla, \dot{\Delta}_j] a.$$

$$C_j(x) = \int h_j(x-y)[v(x) - v(y)] \cdot \nabla a(y)\, dy.$$

The velocity difference contributes $|x-y|\|\nabla v\|_\infty$, and $\||x|h_j\|_1 = 2^{-j}\||x|h\|_1$.

After Bony decomposition and frequency summation

For $1 < p < \infty$, $0 < s < 1 + d/p$,

$$\boxed{\sum_j 2^{js}\|C_j\|_p \lesssim \|\nabla v\|_{\dot{B}^{d/p}_{p,1}} \|a\|_{\dot{B}^{s}_{p,1}}.}$$

The kernel displays the cancellation. Bony handles the other interactions. A bare Lipschitz bound does not propagate arbitrary high regularity.

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Besov estimates for nonlinear transport

For $1 < p < \infty$, $0 < s < 1 + d/p$, define

$$V(T) = \int_0^T \|\nabla v(t)\|_{\dot{B}^{d/p}_{p,1}}\, dt.$$

The blockwise L^p estimate, the commutator bound, and Gronwall give

$$\boxed{\|a\|_{\widetilde{L}^\infty_T \dot{B}^s_{p,1}} \le e^{CV(T)} \left(\|a_0\|_{\dot{B}^s_{p,1}} + \|f\|_{L^1_T \dot{B}^s_{p,1}} \right).}$$

When the transporting velocity is also unknown

If $v = u$, control $V(T)$ through the velocity equation:

$$V(T) \lesssim \|u\|_{L^1_T \dot{B}^{d/p+1}_{p,1}}.$$

The structure preserves the regularity of a, without giving heat smoothing.

These are a priori estimates for sufficiently smooth solutions with the stated norms. A wider admissible range is recorded in backup B4.

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High and low frequencies of the heat flow

$$\|e^{\nu t\Delta}\dot{\Delta}_j u_0\|_p \le Ce^{-c\nu t 2^{2j}}\|\dot{\Delta}_j u_0\|_p.$$

Fix a threshold j_0. Every fixed annulus has exponential decay.

High frequencies: $j > j_0$

The rates are bounded below by a positive constant depending on ν, j_0.

This gives a **uniform exponential rate** for the high-frequency part.

Low frequencies: $j \le j_0$

$$\nu 2^{2j} \longrightarrow 0 \quad (j \to -\infty).$$

There is **no uniform positive rate** over all low blocks on $\mathbb{R}^d$.

The low-frequency behavior needs its own analysis.
For coupled systems, even the type of damping can change with frequency.

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Hybrid Besov norms

Split the homogeneous blocks at a fixed integer j_0

$$z^\ell = \sum_{j \le j_0} \dot{\Delta}_j z = \dot{S}_{j_0+1} z, \qquad z^h = \sum_{j > j_0} \dot{\Delta}_j z = z - z^\ell.$$

These are smooth cutoffs. Neighboring blocks overlap.

A different weight in each frequency regime

$$\boxed{\|z\|_{\mathrm{hyb}} = \sum_{j \le j_0} 2^{j s_\ell} \|\dot{\Delta}_j z\|_{p_\ell} + \sum_{j > j_0} 2^{j s_h} \|\dot{\Delta}_j z\|_{p_h}.}$$

The PDE determines the exponents and the threshold.
Different unknowns can require different hybrid norms.

Compressible coupling will determine the useful hybrid exponents.

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The compressible Navier–Stokes system

Barotropic flow, constant viscosity, and positive density ϱ

$$\begin{cases} \partial_t \varrho + \operatorname{div}(\varrho u) = 0, \\ \varrho(\partial_t u + u \cdot \nabla u) - \mu \Delta u - (\lambda + \mu) \nabla \operatorname{div} u + \nabla P(\varrho) = 0. \end{cases}$$

Assume $\mu > 0$, $\lambda + 2\mu > 0$, and $P'(1) > 0$.

Perturb the equilibrium: $\varrho = 1 + a$, $u = 0$

$$\partial_t a + u \cdot \nabla a = -(1 + a) \operatorname{div} u.$$

Density is transported. Velocity has viscosity and convection.
Pressure and divergence couple the two equations.

The ingredients we have developed now occur in the same system.

Small $\|a\|_{\dot{B}^{d/p}_{p,1}}$ can ensure $1 + a \geq 1/2$. Nonlinear coefficients also require composition estimates for $F(a) - F(0)$.

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Two frequency regimes in compressible flow

Linearization at $(\varrho, u) = (1, 0)$, with $c^2 = P'(1) > 0$

$$\begin{cases} \partial_t a + \operatorname{div} u = 0, \\ \partial_t u - \mu \Delta u - (\lambda + \mu) \nabla \operatorname{div} u + c^2 \nabla a = 0. \end{cases}$$

Low frequencies

Damped acoustic waves.

Density and longitudinal velocity are estimated together.

High frequencies

A parabolic mode and a damped mode coexist.

Density does not acquire free heat smoothing.

Hybrid regularity reflects these distinct spectral mechanisms.

The coefficient-dependent eigenvalues and their asymptotics are in backup B6.

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A glimpse of the time-periodic setting

Optional preview: $u(t+\mathcal{T},x)=u(t,x)$

$$u(t,x)=\sum_{n\in\mathbb{Z}}U_n(x)e^{2\pi int/\mathcal{T}},\qquad U_n(x)=\frac{1}{\mathcal{T}}\int_0^{\mathcal{T}}u(t,x)e^{-2\pi int/\mathcal{T}}\,dt.$$

$$\overline{u}=U_0,\qquad u_{\mathrm{osc}}=u-\overline{u}=\sum_{n\neq 0}U_n e^{2\pi int/\mathcal{T}}.$$

Chemin–Lerner

Take time norms of **spatial** dyadic blocks $\dot{\Delta}_j u$.

Besov regularity in time

Apply the dyadic recipe to **temporal** Fourier frequencies $2\pi n/\mathcal{T}$.

Next talk: combine the spatial estimates with a periodic time structure.

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Take-home message and next talk

Besov spaces match the norm to the frequency structure of the PDE.

$$u = \sum_j \dot{\Delta}_j u, \qquad 2^{js}\|\dot{\Delta}_j u\|_p, \qquad \|\cdot\|_{\ell^r}.$$

20 October 2026, 9:00 AM

Time-Periodic Compressible Navier–Stokes Equations in Besov Spaces

Thank you.

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B1. Homogeneous spaces: the convention

$$\mathcal{S}'_h = \{u \in \mathcal{S}' : \|\dot{S}_j u\|_\infty \longrightarrow 0 \text{ as } j \to -\infty\}.$$

For $u \in \mathcal{S}'_h$,

$$\sum_{j=m}^{n} \dot{\Delta}_j u = \dot{S}_{n+1} u - \dot{S}_m u \longrightarrow u \quad \text{in } \mathcal{S}'.$$

Completeness and PDE intersections

This realization is Banach when $s < d/p$, or $s = d/p$ with $r = 1$.

A higher norm such as $\dot{B}^{d/p+1}_{p,1}$ is used together with a lower normalized space, for example

$$u \in \widetilde{L}^\infty_T \dot{B}^{d/p-1}_{p,1} \cap \widetilde{L}^1_T \dot{B}^{d/p+1}_{p,1}.$$

Danchin (2015), Sections 2.1–2.2. High regularity alone does not fix low-frequency reconstruction.

B2. Why the endpoint needs $r = 1$

Choose a Schwartz function ψ with annular Fourier support and $\psi(0) > 0$.
Take widely separated indices $j_k = 4k$ and define

$$u_N(x) = N^{-1/r} \sum_{k=1}^{N} \psi(2^{j_k} x).$$

Each dilate meets a fixed number of dyadic blocks, independent of N.

$$\|\psi(2^{j_k}\cdot)\|_p = 2^{-j_k d/p} \|\psi\|_p, \qquad \|u_N\|_{\dot{B}^{d/p}_{p,r}} \lesssim 1.$$

But

$$u_N(0) = \psi(0) N^{1-1/r} \longrightarrow \infty \qquad (r > 1).$$

For $r = \infty$, use $N^{-1/r} = 1$. Each u_N is Schwartz, so this disproves a uniform endpoint embedding on the whole space.

B3. Where the tame condition $s > 0$ enters

Paraproduct: only neighboring output indices

$$\|\dot{\Delta}_j T_u v\|_p \lesssim \|u\|_\infty \sum_{|k-j|\le N_0} \|\dot{\Delta}_k v\|_p.$$

The 2^{js} weights are comparable in this finite sum.

Remainder: high inputs can have low output

$$2^{js}\|\dot{\Delta}_j R(u,v)\|_p \lesssim \|u\|_\infty \sum_{k\ge j-N_0} 2^{(j-k)s}(2^{ks}\|\widetilde{\dot{\Delta}}_k v\|_p).$$

The sequence 2^{-ms} for $m \ge -N_0$ is summable precisely when $s > 0$.

Discrete Young's inequality completes this estimate.
A separate endpoint argument is needed if $s = 0$.

B4. A precise transport estimate

Take $1 < p < \infty$, $1 \le r \le \infty$, $p' = p/(p-1)$ and

$$-\min(d/p, d/p') < s < 1 + d/p.$$

For a smooth solution of $\partial_t a + v \cdot \nabla a = f$, set

$$V(t) = \int_0^t \|\nabla v(\tau)\|_{\dot{B}^{d/p}_{p,1}} \, d\tau.$$

$$\boxed{\|a\|_{\widetilde{L}^\infty_t \dot{B}^s_{p,r}} \le e^{CV(t)} \left(\|a_0\|_{\dot{B}^s_{p,r}} + \|f\|_{\widetilde{L}^1_t \dot{B}^s_{p,r}} \right).}$$

The estimate also holds at $s = 1 + d/p$ when $r = 1$.

Danchin (2015), Theorem 2.2 and equation (24), with $p_1 = p$. The displayed norm is stronger than $\dot{B}^{d/p}_{p,\infty} \cap L^\infty$. This is an a priori estimate with the homogeneous convention.

B5. A check on the mixed-norm direction

Let $b_1 = \mathbf{1}_{(0,1)}$ and $b_2 = \mathbf{1}_{(1,2)}$.
Two model dyadic amplitudes peak at different times.

Sequence exponent $r = 1$, time exponent $\rho = \infty$

Bochner	Chemin–Lerner
$\sup_t(b_1 + b_2) = 1$	$\sup_t b_1 + \sup_t b_2 = 2$

Sequence exponent $r = \infty$, time exponent $\rho = 1$

Bochner	Chemin–Lerner
$\int \max(b_1, b_2)\, dt = 2$	$\max\left(\int b_1, \int b_2\right) = 1$

B6. The acoustic eigenvalues

For $q = |\xi|$ and $\nu_L = \lambda + 2\mu > 0$, the unforced longitudinal modes satisfy

$$\partial_{tt}\hat{a} + \nu_L q^2 \partial_t \hat{a} + c^2 q^2 \hat{a} = 0.$$

$$\sigma_{\pm}(q) = \frac{-\nu_L q^2 \pm \sqrt{\nu_L^2 q^4 - 4c^2 q^2}}{2}.$$

Two different regimes

$$q \to 0: \qquad \sigma_{\pm} = -\tfrac{1}{2}\nu_L q^2 \pm icq + O(q^3).$$

$$q \to \infty: \qquad \sigma_{+} \longrightarrow -c^2/\nu_L, \qquad \sigma_{-} \sim -\nu_L q^2.$$

High frequencies have both a damped mode and a parabolic mode.
The transverse velocity modes decay at rate μq^2.

Direct calculation for the displayed linear system. Compare Danchin (2015), Section 4.1.

B7. References and reading guide

H. Bahouri, J.-Y. Chemin, R. Danchin (2011)
Fourier Analysis and Nonlinear Partial Differential Equations.
Grundlehren der mathematischen Wissenschaften 343, Springer.
Chapter 2: Littlewood–Paley theory.
Chapter 3: transport and transport-diffusion equations.
doi:10.1007/978-3-642-16830-7

R. Danchin (2015), freely accessible survey
Fourier analysis methods for the compressible Navier–Stokes equations.
arXiv:1507.02637.
Propositions 2.1–2.3: Bernstein, embeddings, paraproduct.
Theorem 2.1 and Remark 2.2: heat estimates and time norms.
Theorem 2.2: transport. Section 4.1: acoustic spectral analysis.

B8. The critical multiplier estimate

Set $A_k = 2^{kd/2}\|\dot{\Delta}_k f\|_2$ and $B_k = 2^{k\sigma}\|\dot{\Delta}_k g\|_2$.
Assume $-d/2 < \sigma \le d/2$. Then $A, B \in \ell^1$.

The paraproducts

$T_f g$: use $\|f\|_\infty \lesssim \|A\|_{\ell^1}$. For $T_g f$,

$$2^{k\sigma}\|\dot{S}_{k-1} g\, \dot{\Delta}_k f\|_2 \lesssim A_k \sum_{\ell \le k-2} 2^{(\ell-k)(d/2-\sigma)} B_\ell.$$

The double sum is bounded by $\|A\|_1 \|B\|_1$, also at $\sigma = d/2$.

The remainder

Use $L^2 \cdot L^2 \subset L^1$ and $\|\dot{\Delta}_j w\|_2 \lesssim 2^{jd/2}\|w\|_1$:

$$2^{j\sigma}\|\dot{\Delta}_j R(f,g)\|_2 \lesssim \sum_{k \ge j - N_0} 2^{(j-k)(\sigma+d/2)} A_k \widetilde{B}_k, \quad \widetilde{B}_k = \sum_{|\ell - k| \le 1} B_\ell.$$

Sum in j since $\sigma + d/2 > 0$, then in k.